Acceleration and variability in the growth of children's early vocabulary

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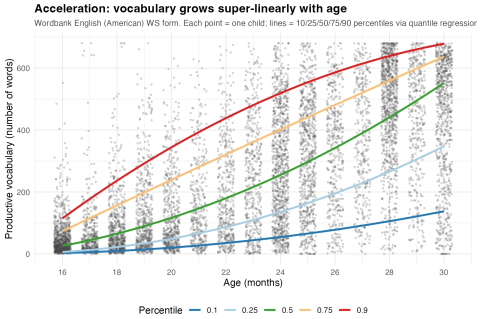
Vocabulary growth

Theoretical: Growth mechanisms are diagnostic of how children's learning works — and how it changes with development.

Practical: Variation in vocabulary growth is a key precursor of academic outcomes.

Methodological: Vocabulary count has a property nearly no other psychological metric has: a true zero.



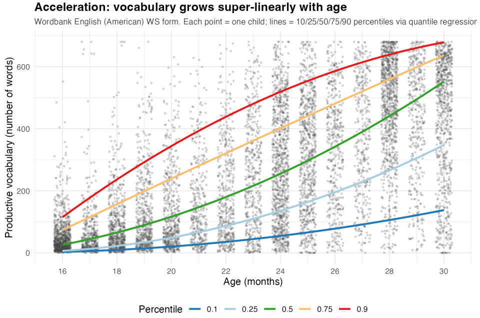
The data

MacArthur–Bates Communicative Development Inventory (CDI) is a parent-report checklist of ~ 680 words. Parents check which words their child *produces* or *comprehends*.

Wordbank: open repository of CDI administrations. ~ 30 languages, $\sim 10^5$ children. We use both:

- **Cross-sectional**: one snapshot per child
- **Longitudinal**: ≥ 2 admins per child, lets us track per-child growth

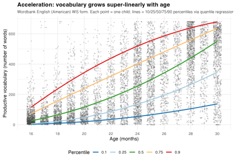
Large N, parent-report calibrated to direct testing, cross-linguistically validated.



One child = one point. Vocabulary × age in English (American) WS.

Two signatures of vocabulary growth

Signature 1: acceleration. Each quantile curve is concave up: doubling age more than doubles vocabulary. The growth rate accelerates. **Signature 2: variability.** The quantile lines fan out: between-child differences in vocabulary grow with age, not shrink.



These signatures are remarkably robust: replicate across languages, samples, and forms. Any model of early word learning must explain both.

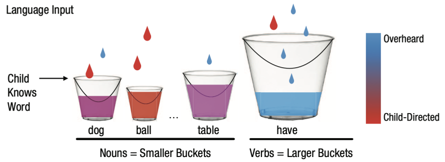
The dominant theoretical view: pure accumulation

Each child has input rate r_i; word j has difficulty δ_j. A word is acquired when enough tokens accumulate.

Textbook lineage:

- McMurray (2007) — pure rate $\times$ time
- Mitchell & McMurray (2009) — + leverage feedback
- Hidaka (2013) — cumulative-Weibull; rate change
- Mollica & Piantadosi (2017) — per-word Gamma
- Kachergis et al. (2021) — IRT Rasch + α_i

All share: between-child variation lives only in baseline (r_i or α_i); per-child scaling exponent fixed.



Each child hears tokens at rate $r_i \cdot p_j$; word j is acquired when its bucket fills past δ_j.

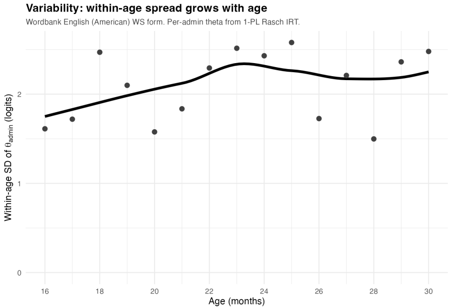
A puzzle: input matters less than supposed

The accumulator view predicts variation in growth comes from variation in input quantity. Coffey & Snedeker (2026) meta-analyzed 71 studies, 4,760 children:

R^2 **of input on outcome = 0.04–0.07.**

Caregiver speech quantity explains 4–7% of variance in language outcome — a real but small effect.

Hypothesis: Variability comes from *how children accelerate*, not how much input they hear.



Between-child SD(θ) grows from $\sim$ 1.7 to $\sim$ 2.3 logits between 16 and 30 mo. If input explains $\sim$ 5%, the rest comes from somewhere.

This work: modeling individual variability in acceleration

We build a Bayesian model to estimate three things directly:

1. **Acceleration.** Is population vocabulary growth super-linear in cumulative input?
2. **Variability.** Do children differ in *how* and *when* their growth accelerates, not just in baseline ability?
3. **Input share.** How much of the between-child variance is attributable to input rate, vs. to internal factors like efficiency?

Does the same machinery applied to large language models yield the same answers? **Spoiler**: no — LLMs show a categorical disanalogy with children's learning. (Later in the talk.

Outline

1. Building the model: individual variability in acceleration
2. Input quantity: how much does it really vary?
3. Results: English and Norwegian longitudinal fits
4. Extensions: directly observed input and processing
5. Connecting to LLMs
6. Synthesis

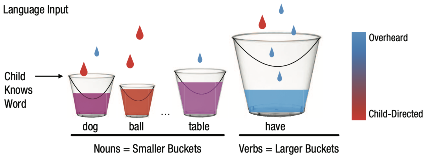
The accumulator hypothesis

The shared idea. Each child accumulates word-evidence at her own input rate r_i; word j is acquired once its difficulty δ_j is crossed.

Ability scales as a power of input. If kids vary only in *rate*, then ability grows as $(\text{tokens})^1$ — the pure *unit accumulator*. Between-child differences come only from differences in r_i.

What our model will check.

- Is the growth exponent really 1, or steeper?
- Are children all alike except for input rate, or do they also differ in efficiency, in growth rate, and in when growth starts?



Bucket-fill metaphor: each child hears tokens at rate $r_i \cdot p_j$; word j is acquired when its bucket fills past difficulty δ_j (Kachergis et al. 2021).

Prior accumulator models: structural lineage

McMurray (2007). Pure accumulator. Each word w has acquisition threshold τ_w from population distribution $g(\tau)$. Acceleration requires $g'(\tau) > 0$.

Mitchell & McMurray (2009). McMurray + leverage feedback $a(t) = t + cL(t)$. *Shifts* growth without creating acceleration.

Hidaka (2013). Closed-form AoA distributions: cumulative (Gamma), rate-change (Weibull, exponent D on time), or both.

Mollica & Piantadosi (2017). Per-word Gamma waiting time: AoA $\sim \Gamma(k_w, \lambda_w)$.

Kachergis et al. (2021). 1-PL Rasch IRT with per-child θ_i and linear age covariate.

Each uses different notation. After we introduce our model, we'll re-express each in our coordinates and compare structurally.

The model: Rasch IRT + accumulator dynamics

$$P(\text{produces}_{ij}) = \sigma(\eta_{ij}) \quad \text{with} \quad \eta_{ij} = \theta_i - \delta_j$$

$$\eta_{ij} = \underbrace{\theta_i}_{\substack{\textbf{kid ability} \\ \textit{(grows with experience)}}} - \underbrace{\delta_j}_{\substack{\textbf{word difficulty} \\ \textit{(per-word, free)}}}$$

This is a standard Rasch IRT model. The substantive content is in how θ_i depends on accumulated experience. We build that up in three steps.

Step 1: pure accumulator

$$\theta_i = \log r_i + \log t \quad \Longleftrightarrow \quad \exp(\theta_i) = r_i \cdot t$$

$$\theta_i \quad = \quad \underset{\substack{\textit{per-child input rate} \\ \textit{(words per month)}}}{\underset{\big\uparrow}{\log r_i}} \quad + \quad \underset{\substack{\textit{time = age} \\ \textit{(months since onset)}}}{\underset{\big\uparrow}{\log t}}$$

Ability = rate × time. The only thing that changes across development is more accumulated input. The only source of variation between children is variation in the input rate r_i.

Step 2: + per-child efficiency α_i

$$\theta_i = \log r_i + \log \alpha_i + \log t \quad \Longleftrightarrow \quad \exp(\theta_i) = \alpha_i \cdot r_i \cdot t$$

$$\theta_i = \underset{\substack{\uparrow \\ \textit{input rate}}}{\log r_i} + \underset{\substack{\uparrow \\ \textbf{\textit{per-child}} \\ \textbf{\textit{efficiency}} \\ \textbf{\textit{(NEW)}}}}{\log \alpha_i} + \underset{\substack{\uparrow \\ \textit{time}}}{\log t}$$

Each child has a multiplicative efficiency α_i over and above their input rate. **The question we'll ask of the fit:** how much of the between-child variance comes from efficiency vs. input rate? We summarize this share as $\pi_\alpha = \sigma^2_\alpha / (\sigma^2_\alpha + \sigma^2_r)$.

Step 3: + population acceleration κ_{pop}

$$\theta_i = \log r_i + \log \alpha_i + \kappa_{\text{pop}} \log t \quad \Longleftrightarrow \quad \exp(\theta_i) = \alpha_i \cdot r_i \cdot t^{\kappa_{\text{pop}}}$$

$$\theta_i \quad = \quad \underset{\substack{\uparrow \\ \textit{input rate}}}{\log r_i} \quad + \quad \underset{\substack{\uparrow \\ \textit{efficiency}}}{\log \alpha_i} \quad + \quad \underset{\substack{\uparrow \\ \textbf{\textit{population}} \\ \textbf{\textit{scaling}} \\ \textbf{\textit{exponent}} \\ \textbf{\textit{(NEW; shared)}}}}{\kappa_{\text{pop}}} \quad \cdot \quad \underset{\substack{\uparrow \\ \textit{time}}}{\log t}$$

The population accelerates: ability scales as $t^{\kappa_{\text{pop}}}$ super-linearly. All children still share one κ_{pop}.
What we'll ask: is κ_{pop} really above 1 — i.e., does ability grow faster than a unit accumulator predicts? This is Hidaka's rate-change model.

Step 4: + per-child acceleration ζ_i

$$\theta_i = \log r_i + \log \alpha_i + (\kappa_{\text{pop}} + \zeta_i) \log t$$

$$\theta_i = \log r_i + \underbrace{\log \alpha_i}_{\textit{efficiency}} + (\underbrace{\kappa_{\text{pop}}}_{\textit{pop. scaling}} + \underbrace{\zeta_i}_{\textbf{\textit{per-child acceleration deviation (NEW)}}}) \cdot \log t$$

Each child has her own growth exponent $\kappa_i = \kappa_{\mathbf{pop}} + \zeta_i$. The new piece: between-child variation isn't just in baseline ability, but in *how trajectories accelerate*. Every prior accumulator model forces every child to share one growth rate. **What we'll ask:** do children really differ in how fast their vocabulary accelerates?

Step 5: + per-child trajectory phase s_i

$$\theta_i = \log r_i + \log \alpha_i + (\kappa_{\text{pop}} + \zeta_i)\,\log(t - s - s_i)$$

$$\theta_i \;=\; \ldots + \;(\kappa_{\text{pop}} + \zeta_i) \cdot \log\big(\,\underset{\substack{\uparrow\\ \textit{calendar age}}}{t} - s - \underset{\substack{\uparrow\\ \textbf{\textit{per-child}}\\ \textbf{\textit{trajectory}}\\ \textbf{\textit{phase (NEW)}}}}{s_i}\,\big)$$

The same kind of individual-variability addition, but on the *time* dimension: children differ in *when* their developmental clock effectively starts. s_i can be positive (later start) or negative (developmentally ahead). **What we'll ask:** does the timing of growth onset really vary across children, alongside how fast they accelerate?

What every prior model pins; what we free

Model	efficiency σ_α	pop. growth κ_{pop}	per-child growth σ_κ	per-child phase σ_s	Per-word δ_j?
McMurray (2007)	0	1	0	0	free, $\delta_j = \log \tau_j$
Mitchell & McMurray (2009)	0	1*	0	0	free, plus leverage c
Hidaka (2013) cumulative	0	1	0	0	free, $\delta_j = \log(N_j/\delta)$
Hidaka (2013) rate-change	0	$D+1$	0	0	free
Mollica & Piantadosi (2017)	0	1	0	0**	free, $\delta_j = \log(k_j/\lambda_j)$
Kachergis et al. (2021)	free	free	0	0	free
Best-fitting model (this work)	**free**	**free**	**free**	**free**	**free**

What's structurally new. Every prior model pins σ_κ and σ_s to zero (same growth rate and onset for all kids). Our model frees both; we find $\sigma_\kappa \approx 3.5$ and $P(\kappa_i > 1) > 0.997$ — the load-bearing additions that let the model fit acceleration and variability *jointly*.

* *Mitchell & McMurray's leverage isn't linear in* $\log t$. ** *Mollica's Gamma encodes a per-word offset, but shared across kids. See appendix for each model translated individually.*

Outline

Where does σ_r come from? The input-rate prior

The kid-level intercept $\xi_i = \log r_i + \log \alpha_i$ has total variance σ_ξ^2 that the data *does* identify. But the split between input rate (σ_r^2) and efficiency (σ_α^2) is not separately identified within the model:

$$\sigma_\xi^2 \ = \ \sigma_r^2 + \sigma_\alpha^2 \quad \Rightarrow \quad \pi_\alpha \ = \ 1 - \frac{\sigma_r^2}{\sigma_\xi^2}$$

We resolve this by pinning σ_r externally — from data quantifying actual variation in child-directed input rates across families. The model then partitions the data-identified σ_ξ^2 into input and efficiency components.

The pin matters for π_α.

- Bigger $\sigma_r \Rightarrow$ more variance assigned to input $\Rightarrow$ lower π_α.
- Pinning σ_r *too low* would overstate efficiency's share; *too high* would mask it.

Strategy. Compile the empirical literature on adult-token rates per child, derive a defensible σ_r, and report a full sensitivity sweep so the headline doesn't hinge on the central pin.

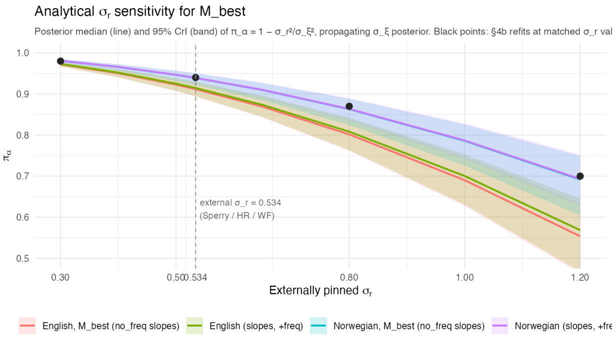
Empirical estimates of input-rate variation

Group	n	Mean (tok/hr)	± 1 SD range (tok/hr)	Tokens/month mean (range)	σ_r (log)
Western, CDS-mother (Sperry sites, HR, WF)	17	$\sim$ 1,100	620–2,000	400K (230K–730K)	0.57
Western, all-adult tokens	9	$\sim$ 2,200	1,450–3,400	800K (530K–1.24M)	0.43
Sperry within-pool, CDS-any-adult	**42**	**$\sim$ 1,500**	**880–2,620**	**547K (320K–956K)**	**0.53**
All groups (incl. Tseltal, Yélî, Pirahã)	27	—	$\sim$ 300–5,000	($\sim$ 110K–1.8M)	1.24

Central pin: $\sigma_r = 0.534$ from the Sperry within-pool. A typical Western child hears $\sim$ 547K adult tokens/month (± 1 SD: $\sim$ 320K–960K, a 2.5$\times$ spread). The Wordbank CDI:WS sample is overwhelmingly WEIRD, so the cross-cultural 1.24 is an outer bound, not a likely value.

Tokens/month = tokens/hr $\times$ H, H = 365 hr/mo (12 waking hr/day $\times$ 30.44 days/mo, model convention).
Sources: Sperry et al. 2019, Hart & Risley 1995, Weisleder & Fernald 2013, Bergelson 2018/2019/2023, Bunce et al. 2024, Casillas 2020/2021, Coffey et al. 2024.

Sensitivity: π_α across plausible σ_r



$\pi_\alpha(\sigma_r) = 1 - \sigma_r^2/\sigma_\xi^2$, with σ_ξ fixed at its data-identified posterior.

Central pin $\sigma_r = 0.534$**:** $\pi_\alpha \approx 0.91 - 0.94$.
Doubled $\sigma_r = 1.2$ **(cross-cultural upper bound):** π_α still > 0.55.
Halved $\sigma_r = 0.30$ **(below any defensible single-site SD):** $\pi_\alpha \approx 0.97$.

Conclusion. The efficiency-dominated decomposition holds across the full range supported by Western input data, and the qualitative answer survives even the cross-cultural upper bound.

Caveat: this is about input *quantity*, not *quality*

What we've modeled. Variation in input *quantity* — how many adult tokens per hour each child hears. Under this lens, between-child differences in vocabulary growth are mostly *not* explained by input quantity.

What we haven't modeled. Variation in input *quality* — diversity, contingency, conversational structure, child-directed register. Quality variation is real and may matter for learning.

Why this is still a defensible scope:

- **Quality is hard to measure at scale.** No cross-population benchmark analogous to LENA-style token rates.
- **Quality would have to be enormous to escape the finding.** A per-child age-varying quality multiplier $q_i(t) = (t/a_0)^{\gamma_i}$ is observationally equivalent to a steeper κ_i — so to absorb the observed acceleration, "quality" would need to grow $\sim 3\{,\}000\times$ for the median kid and $\sim 10^6\times$ for the 95th percentile between 12 and 30 mo. (See appendix.)

Bottom line. Quantity is too small and too well-measured to explain the variability; quality would have to be implausibly large and age-varying to substitute.

Outline

Data and compute for the longitudinal fits

Data.

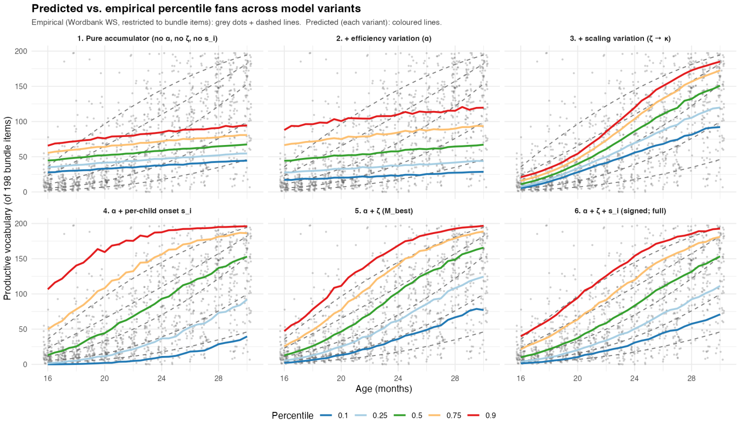
- **English Wordbank (WS form).** $I = 500$ children, ~ 1500 administrations, $J = 671$ items (full WS list), $N \approx 1.1\text{M}$ observed responses. Stratified-by-median-age sampling.
- **Norwegian Wordbank.** $I = 500$ children, ~ 3000 administrations, $J = 716$ items, $N \approx 2.2\text{M}$ responses. Same stratification.

Model family. The same five-step build run as a sequence of Stan models on each dataset: `pure`, $+\alpha$, $+\kappa_{\text{pop}}$, $+\zeta_i$, $+s_i$. Each fit warm-starts from the previous one's posterior.

Compute. Two GCP virtual machines (`n2d-standard-128`, 64 physical AMD EPYC cores each, 128 GB RAM), one per language, running in parallel. Stan with `reduce_sum` threading: 4 chains $\times$ 16 threads/chain saturates the 64 physical cores per machine. $\sim$ 11h per fit, $\sim$ 55h per VM for the five-variant family; the two VMs run in parallel, so $\sim$ 55h wall-clock and $\sim$ 110h of 128-core VM time total. No GPUs.

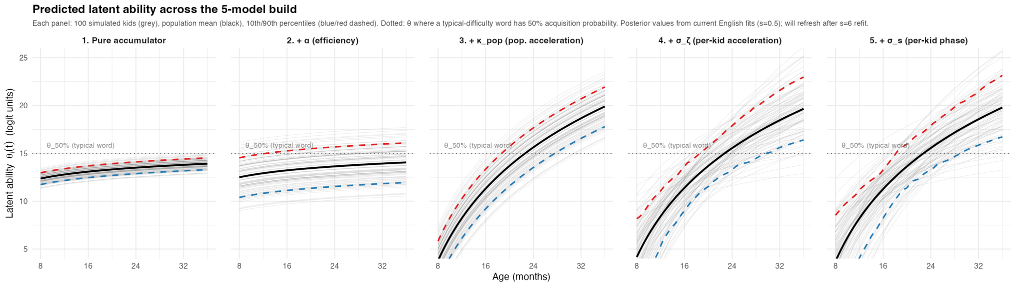
Priors and pins. $s = 6$ mo (population trajectory anchor); $\sigma_r = 0.534$ (Western-CDS empirical pin, see previous slides). Posteriors with $\hat{R} < 1.05$ on all parameters of interest.

Each addition matters: 5-panel comparison to data



Reading. Each panel adds one mechanism. Pure accumulator collapses kids to one trajectory; σ_α adds vertical spread; κ_{pop} adds curvature; σ_ζ opens the fan with age; σ_s handles the floor at young ages.

Same story in θ-space



Same five-panel build, but plotting latent ability $\theta_i(t)$ directly instead of integrating to vocabulary. Population mean (bold), 10th/90th percentiles (blue/red), 100 simulated kids (faint). Dotted: $\theta_{50\%}$, the value at which a typical-difficulty word has 50% acquisition probability.

What it shows. The first two panels never cross $\theta_{50\%}$ within the data range — pure accumulators (with or without efficiency variation) are structurally unable to fit children's acquisition. Each subsequent panel adds a structural mechanism the data demands.

Best-fitting model: parameter posteriors

Parameter	Posterior	What it does
σ_α	1.71 [1.55, 1.91]	Between-child efficiency variation.
$\boldsymbol{\pi_\alpha}$	**0.91 [0.89, 0.93]**	**Share of intercept variance from efficiency, not input.**
$\boldsymbol{\kappa_{\text{pop}}}$	**10.35 [10.21, 10.51]**	**Population scaling exponent: ability scales as (tokens)$^{\kappa_{\text{pop}}}$.**
$\boldsymbol{\sigma_\zeta}$	**3.47 [3.14, 3.84]**	**Between-child variation in scaling exponent.**
$\boldsymbol{\sigma_s}$	**[refits pending]**	**Between-child variation in trajectory phase.**
$\rho(\xi, \zeta)$	$+0.35$ [$+0.22$, $+0.47$]	Higher-baseline kids have steeper growth.

Variability\[0.2em] $\pi_\alpha = 0.91$: efficiency variance dominates input variance.

Acceleration\[0.2em] $\kappa_{\text{pop}} \approx 10.4 \gg 1$: scaling is wildly super-linear, far from the unit accumulator.

Variation in scaling\[0.2em] $\sigma_\kappa > \sigma_\alpha$: kids vary more in scaling exponent than in baseline.

Cross-language acceleration

Sample	admins/kid	$\boldsymbol{\kappa}_{\text{pop}}$	$\boldsymbol{\sigma_\kappa}$	kid range ($\kappa_{\textbf{pop}} \pm 2\sigma_\kappa$)
English	median 3	10.4 [10.2, 10.5]	3.5 [3.1, 3.8]	$\sim$ 3 to 17
Norwegian	median 8	12.4 [12.3, 12.6]	3.7 [3.4, 4.2]	$\sim$ 5 to 20

$\kappa_{\textbf{pop}} \sim 10{-}12$ **in both languages**, with $P(\kappa_i > 1) > 0.997$.

δ **vs.** ζ_i **decomposition is identifiability-dependent.** In Norwegian (8 admins/kid), per-child slopes are well-identified — dropping ζ_i *hurts* fit by 235 ELPD. In English (3 admins/kid), δ pools what Norwegian's data lets fall into ζ_i.

Reading. The acceleration signature is universal. Its internal parameterization is identifiability-dependent on data density.

In any dense longitudinal sample, between-child κ_i *spans roughly* $5{-}15$.

Outline

Three model extensions: testing π_α with richer data

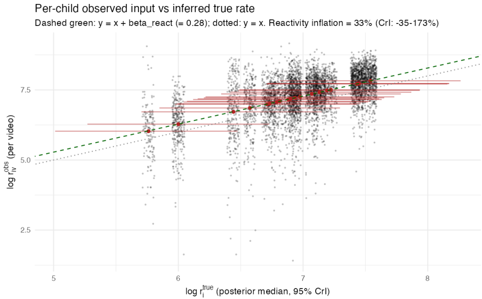
The Wordbank longitudinal fits use *population-level* input rate priors (Sperry/Hart-Risley/Weisleder-Fernald). Two natural follow-ups:

1. **Per-child observed input**: use *measured* per-child input rate instead of a population prior. Does π_α stay ~ 0.9 when input is no longer a pooled estimate? $\rightarrow$ **BabyView, SEEDLingS.**
2. **Per-child processing efficiency**: link CDI growth to an independent per-child readout of processing speed. Does the σ_α we estimate from CDI reflect "processing efficiency" or something else? $\rightarrow$ **Peekbank.**

Both extensions reuse the same model machinery with an added channel in the linear predictor; only the data and the per-child latents change.

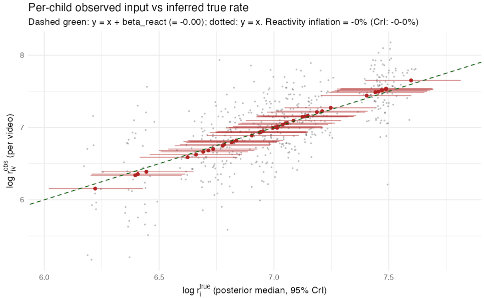
Replicating with directly observed input

BabyView, $N = 20$



$\pi_\alpha = 0.84$,[0.68, 0.93]

SEEDLingS, $N = 44$ **(comp corrected)**



$\pi_\alpha = 0.94$,[0.89, 0.97]

With per-recording observed input, the π_α headline holds: input-rate variance is a small share of between-child variance even when measured directly.

Adding a processing-speed readout: Peekbank LWL

Joint model adds a per-child LWL log-RT readout coupled to $\log \alpha_i$ via γ_{rt}.

Posterior.

- $\gamma_{rt} = 0.082$,$[0.046, 0.125]$ — LWL processing speed is real, and bounded above zero.
- $\pi_\alpha = 0.90$,$[0.86, 0.94]$ — replicates within the Peekbank subset.
- $\rho(\kappa_i, \mathrm{rtslope}_i) = -0.02$,$[-0.33, +0.27]$ — CDI-side and LWL-side growth rates do *not* share variance.

Two clocks, not one. The acceleration in vocab growth and the maturation of processing speed are independent maturation processes within the same child — different "efficiency clocks" ticking in parallel.

This rules out the simplest unification ("one underlying efficiency axis"); the two readouts each pick up a different developmental process.

π_α across five samples

Sample	**N**	$\boldsymbol{\sigma_\alpha}$	$\boldsymbol{\pi_\alpha}$	**Channel**
English longitudinal	200	1.83 [1.65, 2.04]	**0.91 [0.90, 0.93]**	Pooled input prior
Norwegian longitudinal	200	2.10 [1.90, 2.35]	**0.94 [0.93, 0.95]**	Pooled input prior
Peekbank (Stanford)	62	1.64 [1.34, 2.03]	**0.90 [0.86, 0.94]**	Pooled + LWL
BabyView	20	1.13 [0.86, 1.61]	**0.84 [0.68, 0.93]**	Head-mounted video
SEEDLingS	44	1.39 [1.13, 1.74]	**0.94 [0.89, 0.97]**	LENA all-day audio

$\pi_\alpha \in [0.84, 0.94]$ **across all five samples.** The specific input channel (pooled vs. measured) and sample size hardly move the picture: between-child variance is dominated by something *other* than input rate.

(Placeholder: numbers from s=0.5 fits; s=6 refits running. The qualitative pattern is invariant.)

Outline

Why look at LLMs?

LLMs and children both learn word meanings from input. They differ *enormously* in scale and architecture, but the question of how their learning scales with input is statistically well-posed:

- For an LM and a word w, we have $P(w \text{ produced correctly})$ as a function of training tokens.
- This is a sigmoid in log-tokens with a per-word slope.
- The same statistic exists for human children: $P(w \text{ acquired})$ as a function of cumulative input.

This lets us put kids and LMs on the same axis — not just trade anecdotes about scaling. Two questions:

1. Do kids and LMs have similar per-word acquisition slopes?
2. Does a model that fits kids' learning say anything about LMs?

Chang & Bergen (2022): per-word sigmoids in LMs

Setup. Chang & Bergen trained four LMs (BERT, biLSTM, GPT-2, LSTM) on the same input corpus (BookCorpus + WikiText). At each of ~ 200 training checkpoints, they probed each of ~ 611 CDI words to get $P(\text{word produced correctly} \mid \text{checkpoint})$.

For each (LM, word) they fit a 4-parameter logistic

$$S_w(x) = \mathrm{lower}_w + \frac{\mathrm{upper}_w - \mathrm{lower}_w}{1 + \exp((x - \mathrm{mid}_w)/\mathrm{scale}_w)}$$

where $x = \log_{10}(\text{training steps})$, yielding a per-word **scaling slope**: how fast that word's acquisition probability rises with cumulative training. After unit conversion, this is exactly analogous to our $\kappa_i \log t$ for children.

Result they reported. Across the four LMs, mean per-word slope ≈ 1 logit per nat-log training-step. Striking but ambiguous: is this the LM architecture or the input distribution?

Methods: re-running C&B on CHILDES-trained LMs

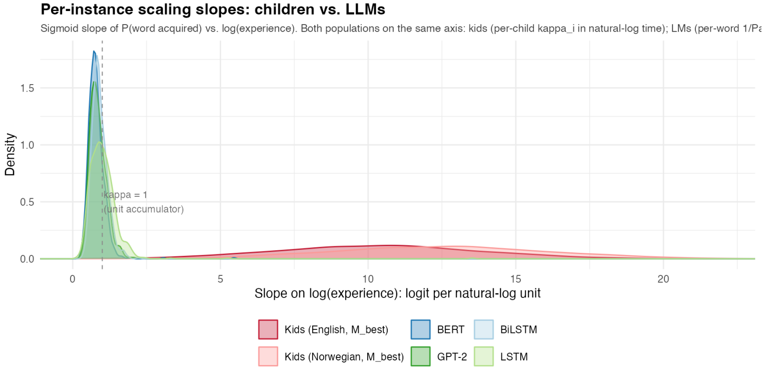
Confound in C&B's setup: BookCorpus + WikiText is adult written text, not child-directed speech. That conflates two things: (a) structural difference in how LMs vs children scale; (b) input distribution mismatch.

Feng et al. (2026) train multiple GPT-2 variants on *CHILDES* — the same corpus that supplies our log p_j frequencies for the kid models. Same architecture as Chang & Bergen's GPT-2, but input matched to what children actually hear.

Our re-application: for each Feng et al. checkpoint, fit C&B's per-word sigmoid on $\sim$ 611 CDI words. Compute per-word slope in our κ-equivalent units. Compare to kids.

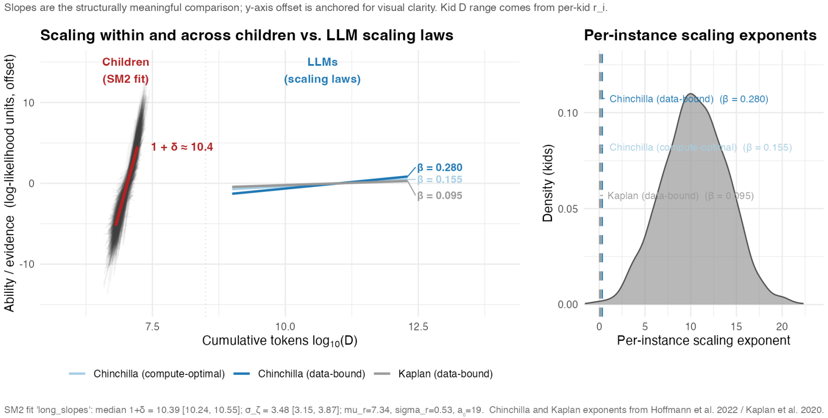
Status: we refit C&B's per-word sigmoids on the Feng et al.
CHILDES-trained GPT-2 checkpoints (Sherlock GPU jobs). The CHILDES-trained slopes are what appear in the comparison on the next slide — not the BookCorpus original.

The per-word view: are kids and LMs learning at the same rate?



Reading. Per-word sigmoid slope on log(experience): LMs (BERT, biLSTM, GPT-2, LSTM, from Chang & Bergen 2022 + our CHILDES-trained refits) vs. our per-child κ_i. **LMs cluster near $\kappa \approx 1$; children average $\sim 10\times$ steeper, with much more between-instance variance.**

The aggregate view: scaling laws



Comparing two estimates of *ability gained per token of experience*: LM scaling laws (Kaplan, Chinchilla compute-optimal, Chinchilla data-bound) all have exponents $\beta \in [0.10, 0.28]$. Children's $\kappa_{\text{pop}} \approx 10$ is two orders of magnitude steeper — a different scaling regime, not a quantitative degree.

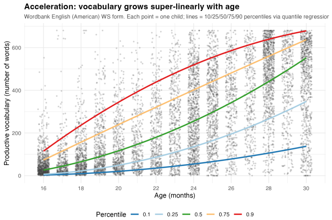
Outline

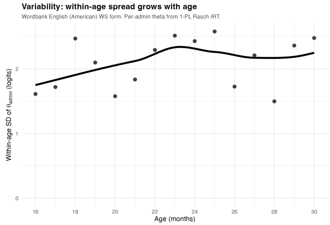
Technical summary

- **Acceleration is real and population-level.** $\{$*(placeholder for s=6 refits)*$\}$ $\kappa_{\text{pop}} \approx 10$, confidently above the unit-accumulator boundary $\kappa_{\text{pop}} = 1$. The first prior model to free κ_{pop} (Hidaka rate-change) was structurally right.
- **Individual variability in acceleration is real too.** $\sigma_\zeta > 0$ and $\sigma_s > 0$: children differ not just in baseline ability, but in *how their trajectories accelerate* and *when*. No prior accumulator model frees these.
- **Input quantity is a small share of between-child variance.** $\pi_\alpha \approx 0.91$ across 5 samples. Aligns with the meta-analytic $R^2 = 0.04$–0.07 of Coffey & Snedeker (2026).
- **LLMs are structurally different.** Same per-word sigmoid statistic gives $\kappa \sim 1$ for LMs vs. ~ 10 for children — a $10\times$ gap with near-zero between-instance variance in LMs.

`github.com/langcog/standard-model-2`

Two signatures, twin constraints on theory





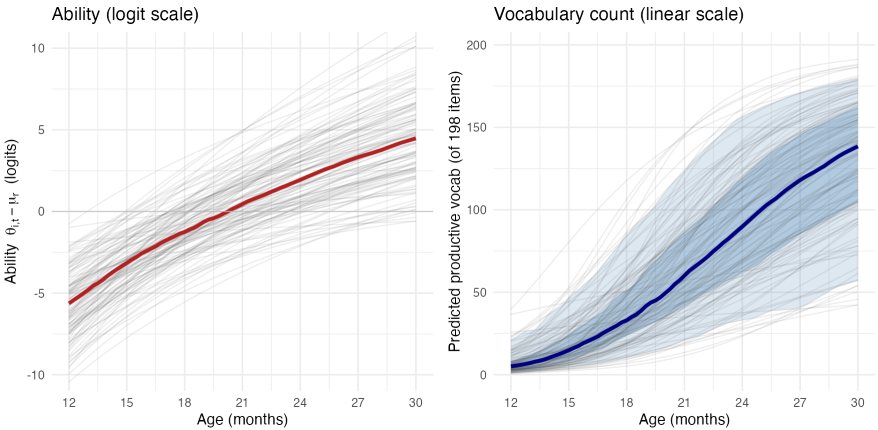
Acceleration. $\kappa_{\text{pop}} \approx 10$ — above any pure accumulator. *Where does the super-linearity come from?*

Variability. $\pi_\alpha \approx 0.91$: input explains $\sim 10\%$ of between-child variance. *Where does the other 90% live?*

Twin constraints on any theory: **(a)** population-level super-linear growth (maturation? learning-to-learn?); **(b)** individual differences *not* in input quantity (processing? interactional quality?). *Pure accumulators satisfy neither.*

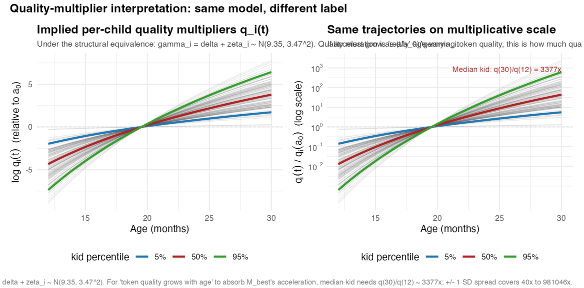
Appendix

Appendix: per-child trajectories from the best fit



120 kids sampled from the best-fitting model's posterior. **Left:** latent ability θ (logit). **Right:** predicted vocabulary. Concave-up shape = super-linear signature; fan width = between-child variability.

Appendix: σ_r sensitivity from earlier cross-sectional fits



Reviewer reflex. "Maybe kids' tokens differ in *quality*, and that's what looks like acceleration."

Structural answer. Per-child age-varying quality $q_i(t) = (t/a_0)^{\gamma_i}$ on input is observationally equivalent to $\gamma_i = \kappa_i - 1$.

Implication. Quality cannot escape acceleration — it *relabels* it. For the median kid, "quality" would have to grow $\sim$ 3,000$\times$ from age 12 to 30 mo. For the 95th percentile kid, $\sim 10^6\times$. Implausible as a substantive quality story.

Appendix: McMurray (2007) in our notation

Original. Each word w has threshold $\tau_w \sim g(\tau)$; deterministic step at $t \geq \tau_w$. Vocabulary $L(t) = G(t)$ (CDF of g). Acceleration requires $g'(\tau) > 0$ over the productive window.

In our notation:

$$\eta_{ij}(t) \;=\; \log t - \log \tau_j \;=\; \kappa \log t - \delta_j \quad \text{with} \quad \kappa \equiv 1, \quad \delta_j = \log \tau_j.$$

Parameter	Setting in McMurray
$\log r_i$	$\equiv 0$ (no per-child rate variation)
$\log \alpha_i$	$\equiv 0$ (no efficiency variation)
$\boldsymbol{\kappa_i}$	$\boldsymbol{\equiv 1}$ (unit accumulator)
δ_j	free, drawn from g

Appendix: Hidaka (2013) in our notation

Original. Three families: cumulative (N-accumulator at constant rate, AoA $\sim \Gamma$), rate-change (1-accumulator at rate $\delta \cdot t^D$, AoA $\sim$ Weibull), and both. **Rate-change**: ability $\propto t^{D+1}$; $D > 0$ produces super-linear scaling.

In our notation (rate-change form):

$$\eta_{ij}(t) \;=\; (D+1)\log t - \delta_j \quad \text{with} \quad \kappa_i \equiv D+1.$$

Parameter	Setting in Hidaka rate-change
$\log r_i$	$\equiv \log \delta_{\text{const}}$ (shared population rate)
$\log \alpha_i$	$\equiv 0$ (no efficiency variation)
$\boldsymbol{\kappa_i}$	$\boldsymbol{\equiv D+1}$ (shared, no per-child variation)
δ_j	free

Hidaka's $D = \kappa_{\mathbf{pop}} - 1$ **in our notation.** His $D > 0$ is our $\kappa_{\text{pop}} > 1$.

Appendix: Mollica & Piantadosi (2017) in our notation

Original. Per-word: k_w effective learning instances arrive as Poisson at rate λ_w; word acquired after k_w events. $\mathrm{AoA}_w \sim \Gamma(k_w, \lambda_w)$. For large k_w, Gamma CDF approximates Gaussian (mean k_w/λ_w) and then a $\sim$ 1.7-scale logistic.

In our notation:

$$\eta_{ij}(t) \;=\; \log t - \delta_j \quad \text{with} \quad \kappa_i \equiv 1,\ \delta_j \approx \log(k_j/\lambda_j).$$

Parameter	Setting in Mollica & Piantadosi
$\log r_i$	$\equiv \log\lambda$ (shared population rate)
$\log\alpha_i$	$\equiv 0$ (no per-child variation)
$\boldsymbol{\kappa_i}$	$\boldsymbol{\equiv 1}$ (unit-rate accumulator)
δ_j	$\approx \log(k_j/\lambda_j)$, free per-word

Appendix: Kachergis et al. (2021) in our notation

Original. 1-PL Rasch IRT with per-child latent ability:
$\mathrm{logit}\, P(y_{ij} = 1) = \theta_i + d_j + \beta_a a_i + \beta_t \log(p_j H a_i)$. Per-child θ_i free; d_j per-word free; shared age slope β_a across all kids.

In our notation:

$$\eta_{ij} \;=\; (\log r_i + \log \alpha_i) + \kappa \log a_i - \delta_j \quad \text{with} \quad \kappa \equiv \text{shared population slope.}$$

Parameter	Setting in Kachergis 2021
$\log r_i$	free, no per-child variance specified
$\boldsymbol{\log \alpha_i}$	**free** (their θ_i absorbs $\log r + \log \alpha$)
κ_i	$\equiv$ shared population slope (no per-child variance)
δ_j	free (their d_j)

First prior model to free σ_α. But σ_κ is still zero — every kid has the same scaling slope.